Results in Control and Optimization 4 (2021) 100022

Contents lists available at ScienceDirect

Results in Control and Optimization

journal homepage: www.elsevier.com/locate/rico

Testing of linear models for optimal control of second-order dynamical system based on model-reality differences

Sie Long Kek [a,*], Sy Yi Sim [b], Chuei Yee Chen [c]

[a] Department of Mathematics and Statistics, Universiti Tun Hussein Onn Malaysia, Pagoh Campus, 84600 Pagoh, Johor, Malaysia
[b] Department of Electrical Engineering Technology, Universiti Tun Hussein Onn Malaysia, Pagoh Campus, 84600 Pagoh, Johor, Malaysia
[c] Department of Mathematics and Statistics, Universiti Putra Malaysia, Serdang, 43400 Seri Kembangan, Selangor, Malaysia

ARTICLE INFO

Keywords:
Conjugate gradient
Linear optimal control
Iterative solution
Adjusted parameters
Model-reality differences

ABSTRACT

In this paper, the testing of linear models with different parameter values is conducted for solving the optimal control problem of a second-order dynamical system. The purpose of this testing is to provide the solution with the same structure but different parameter values in the model used. For doing so, the adjusted parameters are added to each model in order to measure the differences between the model used and the plant dynamics. On this basis, an expanded optimal control problem, which combines system optimization and parameter estimation, is introduced. Then, the Hamiltonian function is defined and a set of the necessary conditions is derived. Consequently, a modified model-based optimal control problem has resulted. Follow from this, an equivalent optimization problem without constraints is formulated. During the calculation procedure, the conjugate gradient algorithm is employed to solve the optimization problem, in turn, to update the adjusted parameters repeatedly for obtaining the optimal solution of the model used. Within a given tolerance, the iterative solution of the model used approximates the correct optimal solution of the original linear optimal control problem despite model-reality differences. The results obtained show the applicability of models with the same structures and different parameter values for solving the original linear optimal control problem. In conclusion, the efficiency of the approach proposed is highly verified.

1. Introduction

Optimal control of a dynamical system, especially in engineering and applied mathematics, is an interesting problem. The dynamic characteristic of a system changing over a time period shows the challenge for determining the solution of a system. Moreover, a dynamical system, which is represented by a class of differential equations [1–3], reveals the real-world applications, and the solution from a dynamical system is eagerly required. In this situation, mathematical modeling and optimal control of the dynamical system could be taken as useful techniques to play a prominent role in solving the complex problems practically. See some examples, brain tumor [4], corruption [5], electoral behavior [6] and Hantavirus infection [7].

In literature, the methodology of solving an optimal control problem with different structures in order to approximate the optimal solution of the original optimal control problem is known as the dynamic integrated system optimization and parameter estimation (DISOPE) algorithm [8,9]. In fact, the DISOPE algorithm was extended from the ISOPE algorithm [10,11], which is an alternative method for solving the steady-state optimization problem. Furthermore, the integrated optimal control and parameter estimation (IOCPE) algorithm has been developed based on the principle of model-reality to handle both discrete-time deterministic and stochastic optimal control problems [12,13]. The basic feature of these algorithms is having the iterative solution leading to

* Corresponding author.
E-mail addresses: slkek@uthm.edu.my (S.L. Kek), sysim@uthm.edu.my (S.Y. Sim), cychen@upm.edu.my (C.Y. Chen).

https://doi.org/10.1016/j.rico.2021.100022
Received 30 September 2020; Received in revised form 2 April 2021; Accepted 27 May 2021
Available online 3 June 2021
2666-7207/© 2021 The Author(s). Published by Elsevier B.V. This is an open access article under the CC BY license (http://creativecommons.org/licenses/by/4.0/).

the true optimal solution using different model structures, where the principle of model-reality differences is satisfied. Furthermore, the applications of these algorithms have been well-defined in the literature. See for some examples in [14–17].

On the other hand, the conjugate gradient approach is an efficient method for solving unconstrained optimization problems [18], it has been applied to solve the optimal control problems [19–21]. In addition to this, incorporating the conjugate gradient approach into the DISOPE algorithm to solve the nonlinear optimal control problems has been studied in [22], where the result presented was quite satisfied. As the model used has a different structure from the real problem, this feature inspires the motivation to look for some outcomes for the same model structure with different parameter values. For this purpose, there are five different linear models that are selected for solving the original optimal control problem of a second-order dynamical system. The calculation procedure of the DISOPE algorithm with the conjugate gradient method is employed, and the iterative results are observed. It is important to emphasize that referring to the study on this paper, the results obtained could lead us to explore the uncertain optimal control problems, which have unknown parameters.

The paper has the following outline. In Section 2, the original linear optimal control problem is described and another linear optimal control model with adjustable parameters is expressed. In Section 3, the system optimization, which incorporates the parameter estimation, is discussed. Then, an equivalent unconstrained optimization problem is formulated so that the conjugate gradient approach is applied for solving the modified optimal control problem. In Section 4, optimal control of a second-order dynamical system is illustrated, where five different models are used for testing the efficiency of the approach proposed. Finally, some concluding remarks are made.

2. Problem description

Consider a linear optimal control problem, given by

$$\begin{aligned}&\min_{u(k)} J_p(u) = \frac{1}{2}x(N)^{\mathrm{T}}S_p(N)x(N) + \frac{1}{2}\sum_{k=0}^{N-1}(x(k)^{\mathrm{T}}Q_p x(k) + u(k)^{\mathrm{T}}R_p u(k))\\ &\text{subject to}\\ &x(k+1) = A_p x(k) + B_p u(k), x(0) = x_0\end{aligned} \tag{1}$$

where $u(k) \in \Re^m, k = 0, 1, \dots, N-1$, and $x(k) \in \Re^n, k = 0, 1, \dots, N$, are, respectively, control sequences and state sequences, whereas A_p is an $n \times n$ transition matrix, B_p is an $n \times m$ control coefficient matrix, $S_p(N)$ and Q_p are $n \times n$ positive semi-definite matrices, and R_p is an $m \times m$ positive definite matrix. Here, J_p is the scalar cost function.

Let this problem be Problem (P), which is named as the original optimal control problem. Theoretically, the analytical solution of Problem (P) exists and the solution procedure is a standard procedure for the linear optimal control problem.

Consider another linear optimal control problem, given by

$$\begin{aligned}&\min_{u(k)} J_1(u) = \frac{1}{2}x(N)^{\mathrm{T}}S(N)x(N) + \gamma(N)\\ &\qquad + \sum_{k=0}^{N-1}\frac{1}{2}(x(k)^{\mathrm{T}}Qx(k) + u(k)^{\mathrm{T}}Ru(k)) + \gamma(k)\\ &\text{subject to}\\ &x(k+1) = Ax(k) + Bu(k) + \alpha(k), x(0) = x_0\end{aligned} \tag{2}$$

where $\gamma(k), k = 0, 1, \dots, N$, and $\alpha(k), k = 0, 1, \dots, N-1$, are introduced as the adjusted parameters. Here, A is an $n \times n$ transition matrix, B is an $n \times m$ control coefficient matrix, $S(N)$ and Q are $n \times n$ positive semi-definite matrices, and R is an $m \times m$ positive definite matrix, while J_1 is the scalar cost function.

This problem is referred to as Problem (M). Notice that the structures of Problem (M) and Problem (P) are the same, however, their parameters values are different. Therefore, our aim is to discuss using a computation approach to obtain the optimal solution of Problem (P) by solving Problem (M). This computation approach is done iteratively. Also, note that these problems are known as the linear quadratic regulator (LQR) problem, and their solution methods are the standard calculation procedure for their respective optimal solution. However, adding the adjusted parameters to Problem (M), and solving it iteratively would provide the correct optimal solution of Problem (P), in spite of model-reality differences.

3. Using conjugate gradient method

Now, introduce an expanded optimal control problem, which is referred to as Problem (E), given by

$$\begin{aligned}\min_{u(k)} J_2(u) &= \frac{1}{2}x(N)^{\mathrm{T}}S(N)x(N) + \gamma(N)\\ &+ \sum_{k=0}^{N-1}\frac{1}{2}(x(k)^{\mathrm{T}}Qx(k) + u(k)^{\mathrm{T}}Ru(k)) + \gamma(k)\\ &+ \frac{1}{2}r_1\|u(k) - v(k)\|^2 + \frac{1}{2}r_2\|x(k) - z(k)\|^2\end{aligned}$$

$$\begin{aligned}
&\text{subject to} \\
&x(k+1) = Ax(k) + Bu(k) + \alpha(k), x(0) = x_0 \\
&\frac{1}{2}z(N)^{\mathrm{T}}S(N)z(N) + \gamma(N) = \frac{1}{2}z(N)^{\mathrm{T}}S_p(N)z(N) \\
&\frac{1}{2}(z(k)^{\mathrm{T}}Q(k)z(k) + v(k)^{\mathrm{T}}Rv(k)) + \gamma(k) = \frac{1}{2}(z(k)^{\mathrm{T}}Q_p(k)z(k) + v(k)^{\mathrm{T}}R_pv(k)) \\
&Az(k) + Bv(k) + \alpha(k) = A_pz(k) + B_pv(k) \\
&v(k) = u(k) \\
&z(k) = x(k)
\end{aligned} \tag{3}$$

where $v(k) \in \Re^m, k = 0, 1, \dots, N-1$, and $z(k) \in \Re^n$, $k = 0, 1, \dots, N$, are introduced to separate the control sequence and the state sequence in the optimization problem from the respective signals in the parameter estimation problem, and $\|\cdot\|$ denotes the usual Euclidean norm. The terms $\frac{1}{2}r_1\|u(k) - v(k)\|^2$ and $\frac{1}{2}r_2\|x(k) - z(k)\|^2$, where $r_1, r_2 \in \Re$ are introduced to improve convexity and to facilitate convergence of the resulting iterative algorithm. It is important to highlight that the algorithm is designed such that the constraints $v(k) = u(k)$ and $z(k) = x(k)$ are satisfied upon the termination of the iterations, assuming convergence is achieved. The state constraint $z(k)$ and the control constraint $v(k)$ are used for computation of the parameter estimation and the matching scheme, while the corresponding state constraint $x(k)$ and control constraint $u(k)$ are reserved for optimizing the model-based optimal control problem. Hence, in this way, system optimization and parameter estimation are mutually interactive.

3.1. Optimality conditions

Define the Hamiltonian function for Problem (E) as follows:

$$\begin{aligned}
H_2(k) =& \frac{1}{2}(x(k)^{\mathrm{T}}Qx(k) + u(k)^{\mathrm{T}}Ru(k)) + \gamma(k) \\
&+ \frac{1}{2}r_1\|u(k) - v(k)\|^2 + \frac{1}{2}r_2\|x(k) - z(k)\|^2 \\
&+ p(k+1)^{\mathrm{T}}(Ax(k) + Bu(k) + \alpha(k)) \\
&- \lambda(k)^{\mathrm{T}}u(k) - \beta(k)^{\mathrm{T}}x(k)
\end{aligned} \tag{4}$$

where $\lambda(k) \in \Re^m, k = 0, 1, \dots, N-1$, and $\beta(k) \in \Re^n, k = 0, 1, \dots, N-1$, are modifiers. Then, the augmented cost function is written by

$$\begin{aligned}
J_2'(u) =& \frac{1}{2}x(N)^{\mathrm{T}}S(N)x(N) + \gamma(N) \\
&+ p(0)^{\mathrm{T}}x(0) - p(N)^{\mathrm{T}}x(N) \\
&+ \xi(N)\left(\frac{1}{2}z(N)^T S_p(N)z(N) - \frac{1}{2}z(N)^T S(N)z(N) - \gamma(N)\right) \\
&+ \Gamma^{\mathrm{T}}(x(N) - z(N)) \\
&+ \sum_{k=0}^{N-1}(H_2(k) - p(k)^{\mathrm{T}}x(k) + \lambda(k)^{\mathrm{T}}v(k) + \beta(k)^{\mathrm{T}}z(k) \\
&+ \xi(k)\left(\frac{1}{2}(z(k)^{\mathrm{T}}Q_pz(k) + v(k)^{\mathrm{T}}R_pv(k)) - \frac{1}{2}(z(k)^{\mathrm{T}}Qz(k) + v(k)^{\mathrm{T}}Rv(k)) - \gamma(k)\right) \\
&+ \mu(k)^{\mathrm{T}}(A_pz(k) + B_pv(k) - Az(k) - Bv(k) - \alpha(k)))
\end{aligned} \tag{5}$$

where $p(k), \xi(k), \mu(k), \Gamma, \lambda(k)$ and $\beta(k)$ are the appropriate modifiers to be determined later.

Using the calculus of variation [1–3,8,12] to (5), the following necessary conditions for optimality are obtained:

(a) Stationary condition:

$$Ru(k) + B^{\mathrm{T}}p(k+1) - \lambda(k) + r_1(u(k) - v(k)) = 0 \tag{6}$$

(b) Co-state equation:

$$p(k) = Qx(k) + A^{\mathrm{T}}p(k+1) - \beta(k) + r_2(x(k) - z(k)) \tag{7}$$

(c) State equation:

$$x(k+1) = Ax(k) + Bu(k) + \alpha(k) \tag{8}$$

(d) Boundary conditions

$$p(N) = S(N)x(N) + \Gamma \quad \text{and} \quad x(0) = x_0 \tag{9}$$

(e) Adjusted parameter equations:

$$\frac{1}{2}z(N)^{\mathrm{T}}S_p(N)z(N) = \frac{1}{2}z(N)^{\mathrm{T}}S(N)z(N) + \gamma(N) \tag{10}$$

$$\frac{1}{2}(z(k)^{\mathrm{T}}Q_p z(k) + v(k)^{\mathrm{T}}R_p v(k)) = \frac{1}{2}(z(k)^{\mathrm{T}}Qz(k) + v(k)^{\mathrm{T}}Rv(k)) + \gamma(k) \tag{11}$$

$$A_p z(k) + B_p v(k) = Az(k) + Bv(k) + \alpha(k) \tag{12}$$

(f) Multiplier equations:

$$\Gamma = (S_p(N) - S(N))z(N) \tag{13}$$

$$\lambda(k) = -(R_p - R)v(k) - (B_p - B)^{\mathrm{T}}\hat{p}(k+1) \tag{14}$$

$$\beta(k) = -(Q_p - Q)z(k) - (A_p - A)^{\mathrm{T}}\hat{p}(k+1) \tag{15}$$

with $\xi(k) = 1$ and $\mu(k) = \hat{p}(k+1)$.

(g) Separable variables

$$z(k) = x(k), v(k) = u(k), \hat{p}(k) = p(k) \tag{16}$$

Note that from the necessary conditions above, (10)–(12) defines the parameter estimation problem, and (13)–(15) give the computation of multipliers. In addition to this, (6)–(9) are the optimality for the modified model-based optimal control problem, which is referred to as Problem (MM), is defined as follows:

$$\begin{aligned}\min_{u(k)} J_3(u) = {} & \frac{1}{2}x(N)^{\mathrm{T}}S(N)x(N) + \gamma(N) \\ & + \sum_{k=0}^{N-1}\frac{1}{2}(x(k)^{\mathrm{T}}Qx(k) + u(k)^{\mathrm{T}}Ru(k)) + \gamma(k) \\ & + \frac{1}{2}r_1\|u(k) - v(k)\|^2 + \frac{1}{2}r_2\|x(k) - z(k)\|^2 \\ & - \lambda(k)^{\mathrm{T}}u(k) - \beta(k)^{\mathrm{T}}x(k) \\ & \text{subject to} \\ & x(k+1) = Ax(k) + Bu(k) + \alpha(k), x(0) = x_0\end{aligned} \tag{17}$$

with the specified $\alpha(k)$,$\gamma(k)$,$\lambda(k)$,$\beta(k)$,Γ,$v(k)$ and $z(k)$, where the boundary conditions are given by x_0 and $p(N)$ with the specified multiplier Γ.

3.2. Conjugate gradient algorithm

Now, refer to [22] for using the conjugate gradient method, define Problem (MM) as an equivalent nonlinear unconstrained optimization problem, and let this problem be Problem (Q), given by

$$\underset{u(k)}{\text{minimize}} J_3(u) \tag{18}$$

with the initial values $u^{(0)} = u(k) \in \Re^m$, $k = 0, 1, \ldots, N-1$. The first-order necessary condition is found to be

$$Ru(k) + B^{\mathrm{T}}p(k+1) - \lambda(k) + r_1(u(k) - v(k)) = 0. \tag{19}$$

Notice that (19) is actually the stationary condition (6) in Problem (MM). In addition, the state and costate variables are updated by (7) and (8), respectively, where the state equation (8) is solved forward and the costate equation (7) is solved backward for the given initial state and the boundary costate in (9).

Denote the gradient of the objective function (18) by

$$\begin{aligned} g(u^{(i)}) &= \nabla J_3(u^{(i)}) \\ &= Ru^{(i)} + B^{\mathrm{T}}p(k+1) - \lambda(k) + r_1(u^{(i)} - v(k)) \end{aligned} \tag{20}$$

where the corresponding control sequence $u^{(i)} = u(k)^i$, and i is the iteration number. Thus, the initial gradient and the initial direction are calculated, respectively, from

$$g^{(0)} = g(u^{(0)}) \tag{21}$$

$$d^{(0)} = -g^{(0)} \tag{22}$$

for an arbitrary initial control sequence $u^{(0)} = u(k)^{(0)}$. While, the control sequence $u^{(i)}$ at each iterative number i is updated from

$$u^{(i+1)} = u^{(i)} + a_i \cdot d^{(i)} \tag{23}$$

where the parameter $a_i \in \Re$ is the step size, which is determined from solving the one-dimensional optimization problem defined as follows:

$$a_i = \underset{a \geq 0}{\text{argmin}} J_3(u^{(i)} + a \cdot d^{(i)}). \tag{24}$$

Here, the term argmin refers to the argument of the minimization of the objective function. Other than this, the new gradient and the new direction will be evaluated from

$$g^{(i+1)} = g(u^{(i+1)}) \tag{25}$$

$$d^{(i+1)} = -g^{(i+1)} + b_i \cdot d^{(i)} \tag{26}$$

based on the new control sequence $u^{(i+1)}$ from (23), where

$$b_i = \frac{g^{(i+1)\mathrm{T}} g^{(i+1)}}{g^{(i)\mathrm{T}} g^{(i)}} \tag{27}$$

is the step size of the search direction $d^{(i+1)}$ in (26) with iteration number $i = 0, 1, 2, \ldots, n$.

Therefore, the result obtained from the conjugate gradient algorithm is presented as a proposition [22] given below:

Proposition 1. *Consider Problem (Q). The control sequence $u^{(i)}$ represented by*

$$u^{(i)} = [(u(0))^{\mathrm{T}}, (u(1))^{\mathrm{T}}, \ldots, (u(N-1))^{\mathrm{T}}],$$

which is defined in (23), is computed by using a set of the direction vectors $d^{(i)}$ whose components are linearly independent. Also, the direction $d^{(i)}$ is conjugacy.

Hence, the conjugate gradient algorithm is summarized in Algorithm 1.

Algorithm 1 (*Conjugate Gradient Algorithm*)**.**

Data Choose an arbitrary initial control sequence $u^{(0)}$.
Step 0 Compute the initial gradient $g^{(0)}$ from (21), and the initial direction $d^{(0)}$ from (22), respectively. Set $i = 0$.
Step 1 Solve the state equation (8) forward in time from $k = 0$ to $k = N$ with the initial condition (9) to obtain $x^{(i)} = x(k), k = 0, 1, \ldots, N$.
Step 2 Solve the costate equation (7) backward in time from $k = N$ to $k = 0$ with the boundary condition (9), where $p^{(i)} = p(k)$ is the solution obtained.
Step 3 Evaluate the cost function J_3 from (18).
Step 4 Determine the step size a_i from (24).
Step 5 Update the control $u^{(i+1)}$ from (23).
Step 6 Calculate the step size b_i from (27).
Step 7 Update the gradient $g^{(i+1)}$, and the direction $d^{(i+1)}$, respectively, from (25) and (26). If the gradient $g^{(i+1)} = g^{(i)}$, stop the iteration, else set $i = i + 1$, and go to Step 1.

Remarks.

(a) In Step 0, setting the initial control sequence $u^{(0)}$ is the preliminary step for applying the conjugate gradient algorithm.
(b) In Steps 1, 2 and 3, system optimization problem based on the corresponding control sequence $u^{(i)}$ is solved.
(c) In Steps 4, 5, 6 and 7, the conjugate gradient algorithm is implemented.
(d) In Step 5, the open-loop control sequences are generated.

3.3. Iterative calculation procedure

According to the discussion above, a summary of the calculation procedure for the approach proposed is made in Algorithm 2.

Algorithm 2 (*Iterative Algorithm*)**.**

Data $A, B, Q, R, S(N), N, x_0, r_1, r_2, k_v, k_z, k_p, A_p, B_p, Q_p, R_p, S_p(N)$ are required.
Step 0 Compute a nominal solution. Assume that $\alpha(k) = 0, k = 0, 1, \ldots, N-1$, and $r_1 = r_2 = 0$. Solve Problem (M) defined by (2) to obtain $u(k)^0, k = 0, 1, \ldots, N-1$, and $x(k)^0, p(k)^0$, $k = 0, 1, \ldots, N$. Then, with $\alpha(k) = 0, k = 0, 1, \ldots, N-1, \gamma(k) = 0, k = 0, 1, \ldots, N$, and using r_1, r_2 from the data. Set $i = 0, v(k)^0 = u(k)^0, z(k)^0 = x(k)^0$ and $\hat{p}(k)^0 = p(k)^0$.
Step 1 Compute the parameters $\gamma(k)^i, k = 0, 1, \ldots, N$, and $\alpha(k)^i, k = 0, 1, \ldots, N-1$, from (10)–(12). This is called the parameter estimation step.
Step 2 Compute the modifiers $\Gamma^i, \lambda(k)^i$ and $\beta(k)^i, k = 0, 1, \ldots, N-1$, from (13)–(15).
Step 3 With $\gamma(k)^i, \alpha(k)^i$, $\Gamma^i, \lambda(k)^i, \beta(k)^i, v(k)^i$ and $z(k)^i$, solve Problem (Q) defined by (18) using the conjugate gradient algorithm as presented by Algorithm 1. This is called the system optimization step.
Step 4 Test the convergence and update the optimal solution of Problem (P). In order to provide a mechanism for regulating convergence, a simple relaxation method is employed:

$$v(k)^{i+1} = v(k)^i + k_v(u(k)^i - v(k)^i) \tag{28}$$

$$z(k)^{i+1} = z(k)^i + k_z(x(k)^i - z(k)^i) \tag{29}$$

$$\hat{p}(k)^{i+1} = \hat{p}(k)^i + k_p(p(k)^i - \hat{p}(k)^i) \tag{30}$$

where $k_v, k_z, k_p \in (0, 1]$ are scalar gains. If $v(k)^{i+1} = v(k)^i, k = 0, 1, \ldots, N-1$, and $z(k)^{i+1} = z(k)^i, k = 0, 1, \ldots, N$, within a given tolerance, stop; else set $i = i + 1$, and repeat the procedure starting from Step 1.

Remarks.

(a) In Step 0, calculating the nominal solution can be done through solving the linear quadratic regulator (LQR) optimal control problem, where the feedback gain and the Riccati equation are solved offline.
(b) In Step 1, the adjusted parameters are zero before the iteration begins, and their values are computed from (10)–(12) in which the differences between the real plant and model used are measured during the iterative procedure.
(c) In Step 2, computing the multipliers takes into account the necessary conditions of Problem (P) and Problem (M).
(d) In Step 3, obtaining the new control sequence is effective as the search direction vectors are conjugacy.
(e) In Step 4, the simple relaxation method, given by (28)–(30), is used for the matching scheme.

3.4. Convergence analysis

During the iterative procedure, some assumptions are made as follows:

Assumption 1. The optimal solution of Problem (P) is given by $\{u^*, x^*, p^*\} = \{v, z, \hat{p}\}$, and the iterative solution of Problem (M) and Problem (MM) is represented by $\{u, x, p\}$.

Assumption 2. The following limits exist.

(a) The limit of control sequences exist, that is, $\lim_{i\to\infty}(u(k)^{(i)} - v(k)^{(i)}) = 0$.
(b) The limit of state sequences exist, that is, $\lim_{i\to\infty}(x(k)^{(i)} - z(k)^{(i)}) = 0$.
(c) The limit of costate sequences exist, that is, $\lim_{i\to\infty}(p(k)^{(i)} - \hat{p}(k)^{(i)}) = 0$.

Now, refer to (12), rewrite the adjusted parameter $\alpha(k)^i$ at the iteration number i to be

$$\alpha(k)^i = (A_p - A)z(k)^i + (B_p - B)v(k)^i. \tag{31}$$

Here, (31) is clearly measuring the differences between the model used and real plant dynamics. By substituting (31) into the state equation (8) to have

$$x(k+1)^i = A(x(k)^i - z(k)^i) + B(u(k)^i - v(k)^i) + A_p z(k)^i + B_p v(k)^i. \tag{32}$$

Thus, from Assumption 2, (32) is simplified to be

$$x(k+1)^i = A_p z(k)^i + B_p v(k)^i \tag{33}$$

Taking (14) into (6), the stationary condition becomes

$$\begin{aligned} &R(u(k)^i - v(k)^i) + B^{\mathrm{T}}(p(k+1)^i - \hat{p}(k+1)^i) \\ &\quad + R_p v(k)^i + B_p^{\mathrm{T}}\hat{p}(k+1)^i + r_1(u(k)^i - v(k)^i) = 0 \end{aligned} \tag{34}$$

and yielding the following result,

$$R_p v(k)^i + B_p^{\mathrm{T}}\hat{p}(k+1)^i = 0. \tag{35}$$

Therefore, the following proposition provides the convergence result for the DISOPE algorithm discussed.

Proposition 2. *With Assumptions 1 and 2, the iterative solution of Problem (MM) is the optimal solution of Problem (P), which is,*

$$\{u^*, x^*, p^*\} = \{v, z, \hat{p}\} = \{u, x, p\},$$

in spite of model-reality differences.

Proof. From the discussion above, the proof is shown by (35), which is the stationary condition for Problem (P) and (33), which is the state equation for Problem (P). This completes the proof.

On the other hand, consider (24), the following proposition is valid for using the conjugate gradient algorithm in solving Problem (Q) and the LQR problem.

Proposition 3. *The gradient of the objective function in Problem (Q) and the subspace of feasible direction are orthogonal, which satisfies*

$$g^{(i+1)\mathrm{T}} d^{(i)} = 0. \tag{36}$$

Proof. The one-dimensional optimization mentioned in (24) can be solved as follows,

$$\frac{dJ_3(u^{(i)} + a \cdot d^{(i)})}{da} = \nabla J_3(u^{(i)} + a \cdot d^{(i)})^{\mathrm{T}} d^{(i)} = g^{(i+1)\mathrm{T}} d^{(i)} = 0 \tag{37}$$

which is the first-order necessary condition of Problem (Q), and gives

$$a_i = -\frac{d^{(i)\mathrm{T}} g^{(i)}}{d^{(i)\mathrm{T}} R d^{(i)}}. \tag{38}$$

Consider the updating rule of the control law in (23) and the stationary condition (35), then, the recursion of the gradient is given by

$$g^{(i+1)} = g^{(i)} + a_i \cdot R d^{(i)}. \tag{39}$$

From (39), it is noticed that by substituting (38),

$$d^{(i)\mathrm{T}} g^{(i+1)} = d^{(i)\mathrm{T}} g^{(i)} + a_i \cdot d^{(i)\mathrm{T}} R d^{(i)} = 0. \tag{40}$$

Also for $j < i$,

$$d^{(j)\mathrm{T}} g^{(i+1)} = d^{(j)\mathrm{T}} g^{(i)} + a_i \cdot d^{(j)\mathrm{T}} R d^{(i)}. \tag{41}$$

Here, the first term on the right-hand side of (41) vanishes because of the induction hypothesis, and the second term vanishes by using the orthogonal property of the feasible direction $d^{(i)}$. Hence, the orthogonality of the gradient and the subspace of the feasible direction is proven. This completes the proof.

4. Illustrative example

In this section, the example of optimal control of a second-order dynamical system [23] is discussed. Let the real problem be Problem (P), which is known as the original optimal control problem. While, another LQR model, which is referred to as Problem (M), is being solved by using the approach proposed for obtaining the optimal solution of Problem (P). These two problems are given as follows.

(a) Problem (P)

$$\begin{aligned}
&\min_{u(k)} J_0(u) = \frac{1}{2}(x_1(20))^2 + (x_2(20))^2 + \frac{1}{2}\sum_{k=0}^{19}(x_1(k))^2 + (x_2(k))^2 + (u(k))^2 \\
&\text{subject to} \\
&x_1(k+1) = 2x_1(k) + x_2(k) + u(k) \\
&x_2(k+1) = x_2(k) + u(t) \\
&x_0 = (2 \quad 1)^{\mathrm{T}}
\end{aligned} \tag{42}$$

where the state transition matrix and control coefficient matrix are

$$A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix} \text{ and } B = \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

(b) Problem (M)

$$\begin{aligned}
&\min_{u(k)} J_1(u) = \frac{1}{2}(x_1(20))^2 + (x_2(20))^2 + \frac{1}{2}\sum_{k=0}^{19}(x_1(k))^2 + (x_2(k))^2 + (u(k))^2 \\
&\text{subject to} \\
&x(k+1) = Ax(k) + Bu(k) + \alpha(k) \\
&x(0) = x_0
\end{aligned} \tag{43}$$

In Problem (M), there are five models with different matrices of state transition A and control coefficient matrix B, which are denoted as M_1, M_2, M_3, M_4, and M_5, respectively, as given in Table 1. Notice that these models have a diagonal of zero in the state transition matrix A, which indicates no inter-relation on the same state, while the control coefficient matrix B provides different weights of the control effort to the respective model.

The simulation results for these models are shown in Table 2. It is remarked that these five models have their respective optimal cost, which is the initial cost before the approach proposed is applied. After implementing the iteration calculation procedure, these five models, which have added the adjusted parameters, have iterative solutions that converge to the true optimal solution of Problem (P) despite model-reality differences. From the results obtained, it is shown that the optimal cost of Problem (P), which is 17.430152 units, has been approximated correctly by using the models suggested.

The graphical result for each model without adding the adjusted parameter is, respectively, shown in Figs. 1, 2, 3, 4, and 5. In these figures, the optimal trajectories of control, state and costate for each model are presented. These trajectories move toward

Table 1
Different models in Problem (M).

Model	State transition matrix A	Control coefficient matrix B
M_1	$\begin{pmatrix} 0.0 & 0.0 \\ 1.0 & 0.0 \end{pmatrix}$	$\begin{pmatrix} -1.0 \\ 2.0 \end{pmatrix}$
M_2	$\begin{pmatrix} 0.0 & 1.0 \\ 0.0 & 0.0 \end{pmatrix}$	$\begin{pmatrix} -2.0 \\ -1.0 \end{pmatrix}$
M_3	$\begin{pmatrix} 0.0 & 0.0 \\ -1.0 & 0.0 \end{pmatrix}$	$\begin{pmatrix} 2.0 \\ 1.0 \end{pmatrix}$
M_4	$\begin{pmatrix} 0.0 & 2.0 \\ 0.0 & 0.0 \end{pmatrix}$,	$\begin{pmatrix} 1.0 \\ -1.0 \end{pmatrix}$
M_5	$\begin{pmatrix} 0.0 & 0.0 \\ -2.0 & 0.0 \end{pmatrix}$	$\begin{pmatrix} 0.0 \\ 1.0 \end{pmatrix}$

Table 2
Simulation results.

Model	Initial cost	Final cost	Iteration number	Elapsed time
M_1	3.244563	17.430152	17	0.136013
M_2	2.686141	17.430153	41	0.336155
M_3	4.291288	17.430153	43	0.352063
M_4	4.186141	17.430152	19	0.139843
M_5	6.500000	17.430152	26	0.214934

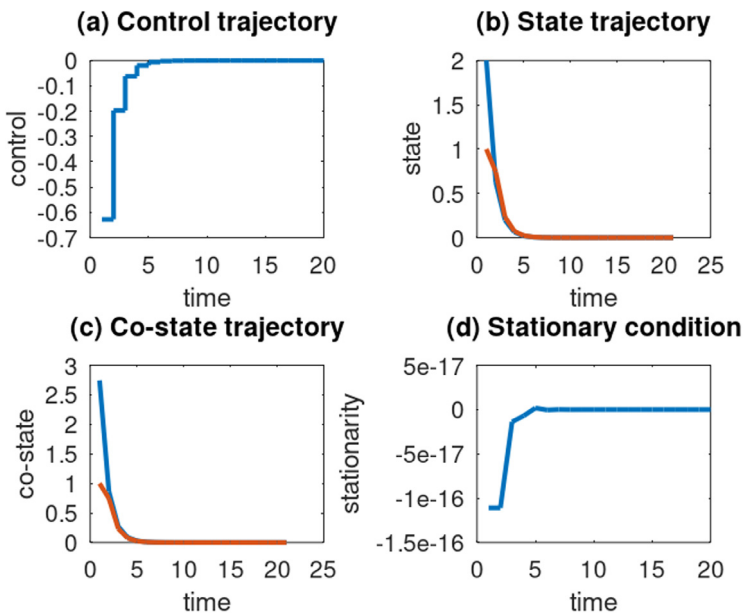

Fig. 1. Problem (M), trajectories for Model M_1 without adjusted parameters.

zero and stay at the steady-state of zero with different transient times. In Fig. 1, the control sequences move from −0.6 toward zero, and stays on zero after 5 units of time, while the state sequences go downward from the initial state and stay on zero after 5 units of time. In Fig. 2, the control sequences, which have the initial value at 0.3, are reducing to zero and staying on zero after 5 units of time. With these control efforts, the state sequences reach the steady-state after 5 units of time. In Fig. 3, the control sequences have the initial value of 0.21 and take about 5 units of time to reach the steady-state of zeros, whereas, the state sequences are also staying on the steady-state of zero after 5 units of time. In Fig. 4, the initial value of the control sequences is −0.3, the steady-state of zero is reached after 5 units of time, which leads the state sequences to reach the steady-state after 5 units of time. In Fig. 5, the control sequences stay on zero after 2 units of time, and the state sequences spend 4 units of time to reach the steady-state of zero.

The given stationary condition in each figure shows that the corresponding model used has the respective optimal solution before using the approach proposed. It is clearly shown that these models used without adding the adjusted parameters have trajectories of

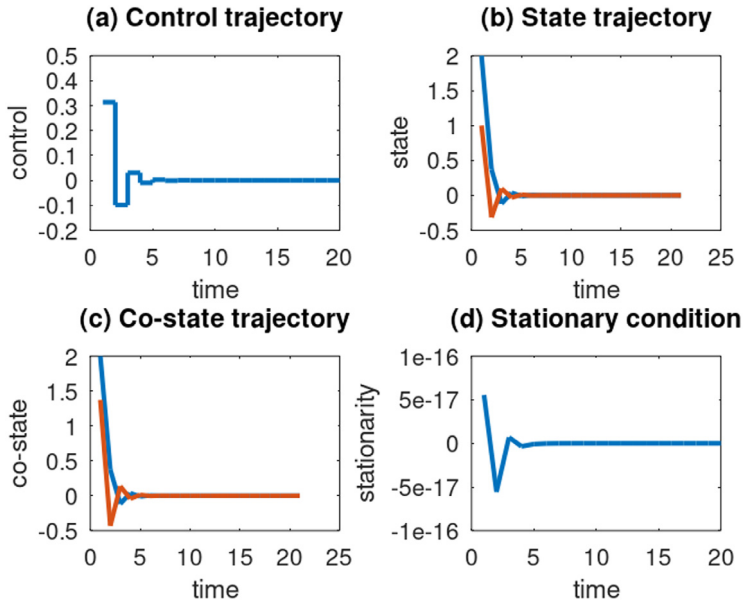

Fig. 2. Problem (M), trajectories for Model M_2 without adjusted parameters.

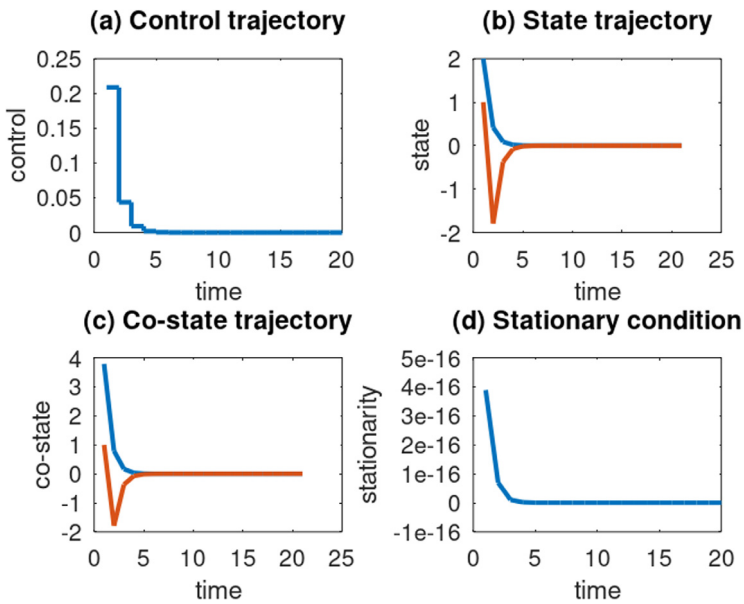

Fig. 3. Problem (M), trajectories for Model M_3 without adjusted parameters.

control and state, which are definitely different from the trajectories of Problem (P) as shown in Fig. 6. Here, the control sequences are moving to the value of 1 from the initial value of −3.5, then turn to be reduced to zeros after 4 units of time. Because of these control efforts, the state sequences are staying at the steady-state of zero after 5 units of time. Nonetheless, trajectories of these models used approximate to the true trajectories of Problem (P), within a given tolerance, at the end of the iteration finally. This approximation can be done successfully due to adding the adjusted parameters into the model used.

From the testing results, the approach proposed in this paper could be a practical method for obtaining the correct optimal solution of the original LQR problem by solving another LQR model when adding the adjusted parameters into the model used. It is highlighted that the original LQR optimal control problem might be unknown or uncertain, however, the output shall be assumed to be measurable. Hence, the approach proposed could be applied properly to provide the approximate solution of the original LQR optimal control problem, in spite of model-reality differences.

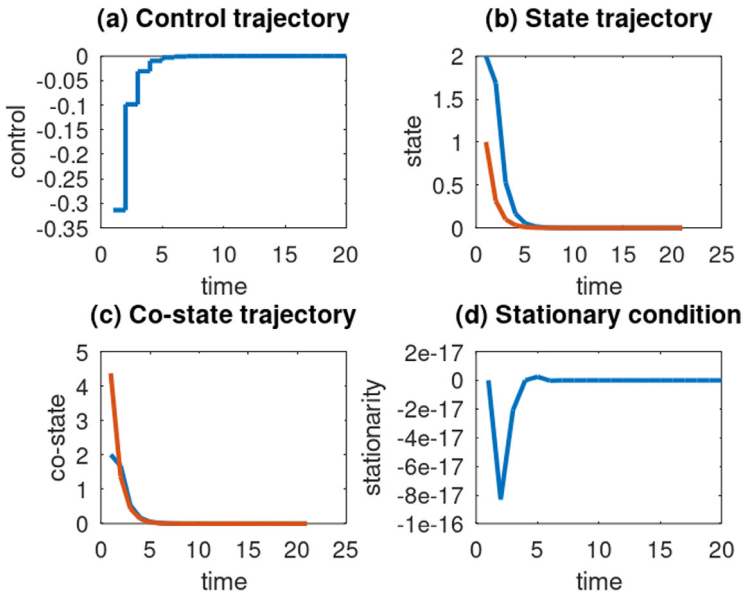

Fig. 4. Problem (M), trajectories for Model M_4 without adjusted parameters.

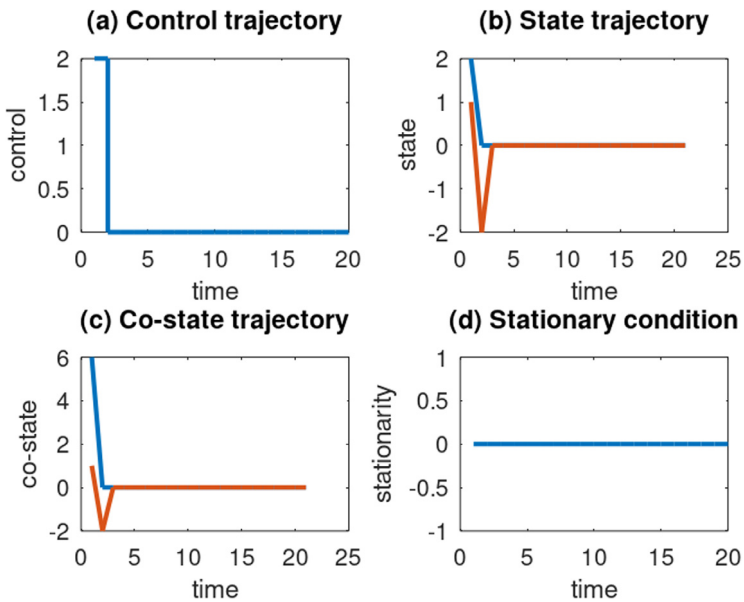

Fig. 5. Problem (M), trajectories for Model M_5 without adjusted parameters.

5. Concluding remarks

The LQR models, which have different parameter values in the state transition and the control coefficient matrices, were discussed in this paper, aiming to approximate the true optimal solution of the original optimal control problem, which has a linear model too. Apparently, the conjugate gradient approach, which associates with the principle of model-reality differences, is employed to solve the LQR models iteratively. This calculation procedure could be done by adding the adjusted parameters into the models, and the differences between the real plant and the models used are measured and updated repeatedly. From the illustration studied, five different LQR models were solved and the correct optimal solution of the original optimal control problem was obtained despite model-reality differences. In conclusion, the efficiency of the approach proposed is highly verified from the testing conducted. For the future research direction, it is suggested to have an efficient approach for identifying the state transition and the control coefficient matrices in the LQR model so that the optimal solution of the nonlinear optimal control problem could be obtained tractably.

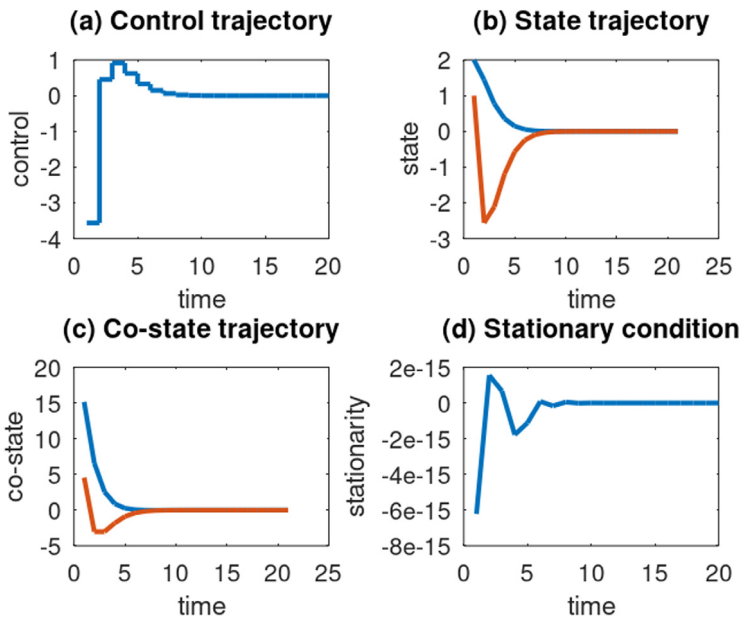

Fig. 6. Problem (P), trajectories for the original model.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The authors would like to thank the Ministry of Education Malaysia (MOE) for supporting this research under Fundamental Research Grant Scheme Vot No. FRGS/1/2018/STG06/UTHM/02/5 and partially sponsored by Universiti Tun Hussein Onn Malaysia (UTHM).

References

[1] Bryson AE, Ho YC. Applied optimal control. Washington, DC, USA: Hemisphere; 1975.
[2] Kirk DE. Optimal control theory: An introduction. Mineola, NY: Dover Publications, New York; 2004.
[3] Lewis FL, Vrabie V, Symos VL. Optimal control. 3rd ed. New York: John Wiley & Sons, Inc.; 2012.
[4] Belmiloudi A. Mathematical modelling and optimal control problems in brain tumor targeted drug delivery strategies. Int J Biomath 2017;10:1759956.
[5] Athithan S, Ghosh M, Li X. Mathematical modelling and optimal control of corruption dynamics. Asian-Eur J Math 2018;11:1850090.
[6] Balatif O, Labzai A, Rachik M. A discrete mathematical modeling and optimal control of the electoral behavior with regard to a political party. 2018, Article ID 9649014.
[7] Yusof FM, Abdullah FA, Md. Ismail AI. Modeling and optimal control on the spread of hantavirus infection. Mathematics 2019;7:1192.
[8] Becerra VM, Roberts PD. Dynamic integrated system optimization and parameter estimation for discrete time optimal control of nonlinear systems. Internat J Control 1996;63:257–81.
[9] Roberts PD, Becerra VM. Optimal control of a class of discrete-continuous nonlinear systems decomposition and hierarchical structure. Automatica 2001;37:1757–69.
[10] Roberts PD. An algorithm for steady-state system optimization and parameter estimation. Internat J Systems Sci 1979;10:719–34.
[11] Roberts PD, Williams TWC. On an algorithm for combined system optimization and parameter estimation. Automatica 1981;17:199–209.
[12] Kek SL, Teo KL, Mohd Ismail AA. An integrated optimal control algorithm for discrete-time nonlinear stochastic system. Internat J Control 2010;83:2536–45.
[13] Kek SL, Mohd Ismail AA, Teo KL, Rohanin A. An iterative algorithm based on model-reality differences for discrete-time nonlinear stochastic optimal control problems. Numer Algebra Control Optim 2013;3:109–25.
[14] Kek SL, Teo KL, Mohd Ismail AA. Filtering solution of nonlinear stochastic optimal control problem in discrete-time with model-reality differences. Numer Algebra Control Optim 2012;2:207–22.
[15] Kek SL, Teo KL, Mohd Ismail AA. Efficient output solution for nonlinear stochastic optimal control problem with model-reality differences. Math Probl Eng 2015;9, Article ID 659506.
[16] Kek SL, Mohd Ismail AA. Output regulation for discrete-time nonlinear stochastic optimal control problems with model-reality differences. Numer Algebra Control Optim 2015;5:275–88.
[17] Kek SL, Li J, Teo KL. Least squares solution for discrete time nonlinear stochastic optimal control problem with model-reality differences. Appl Math 2017;8:1–14.
[18] Chong EKP, Zak SH. An introduction to optimization. 4th ed. Hoboken, New Jersey: John Wiley & Sons, Inc.; 2013.
[19] Lasdon LS, Mitter SK. The conjugate gradient method for optimal control problems. IEEE Trans Automat Control 1967;12:132–8.

[20] Nwaeze E. An extended conjugate gradient method for optimizing continuous-time optimal control problems. Can J Comput Math Nat Sci Eng Med 2011;2:39–44.
[21] Raji RA, Oke MO. Higher-order conjugate gradient method for solving continuous optimal control problems. IOSR J Math 2015;11:88–90.
[22] Kek SL, Sim SY, Leong WJ, Teo KL. Conjugate gradient approach for discrete time optimal control problems with model-reality differences. In: Volosencu Constantin, Saghafinia Ali, Du Xian, Chakrabarty Sohom, editors. Control theory in engineering. IntechOpen; 2019, http://dx.doi.org/10.5772/intechopen.89711, Available from: https://www.intechopen.com/books/control-theory-in-engineering/conjugate-gradient-approach-for-discrete-time-optimal-control-problems-with-model-reality-difference.
[23] Zhang W, Abate A, Hu J. Efficient suboptimal solutions of switched LQR problems. In: 2009 American control conference. 2009, p. 1084–91.